

AntiFraud Solution

AI-Driven Intelligence

Predict, five days in advance, which accounts will cost the B-book money — and act before they do.
A configurable, explainable, self-correcting framework (the TAF engine).

Everything the CEO asked for, on one page.

1

Core metrics & the target

Metric = **Abuse-USD caught in advance** — money Toxic/Arbitrage accounts extract, flagged before they take it. Baseline is measured, not guessed: **\$454,136/day** at stake (perfect-oracle upper bound). **Target: detect $\geq$ 75% out-of-sample.**

Specific metric · **Measurable** in USD & AUC · **Target** set

2

The logic of the system

Observe → **Classify** → **Predict** → **Act** → **Learn**. Transparent rules place each account in measurable classes; an explainable model per class predicts who turns to Abuse in 5 days; the class picks the action. **Every flag traces to a metric & threshold — no black box.**

Explainable end-to-end — the logic can be walked through, step by step

3

The result in 2–3 weeks, vs the target

A **2-week live pilot**, predicted-vs-actual **audited daily** against ground truth. At target it recovers $\geq$ **\$341k/day (~\$86M/yr)** of B-book P&L. **Clear go / no-go.**

Achievable (models already 0.85 AUC) · **Relevant** (core P&L) · **Time-bound** (2 weeks)

The B-book leak has a precise size — and a precise source.

\$454.1k

Addressable abuse per trading day
(perfect oracle)

~\$3.0M

Per week · ~\$114M annualised

98%

of all client winnings flow to
Toxic/Arbitrage accounts

15,598

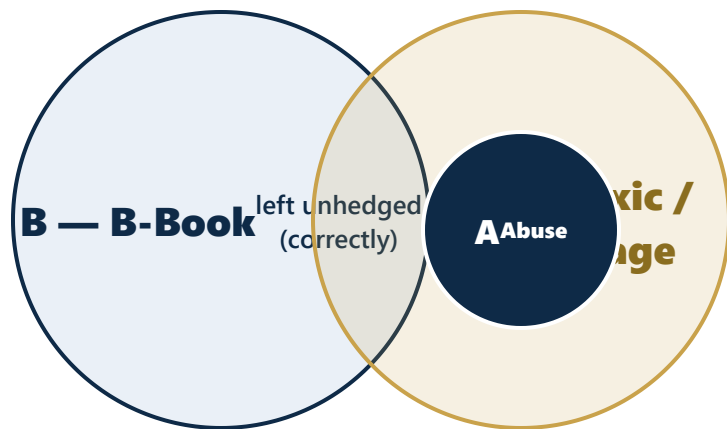
accounts turn abusive in ≥ 1 week (of
36,005)

The accounts that **win** are almost exactly the accounts that are **toxic or arbitraging** by definition.

This is not a coincidence — it is the thesis. If we can identify *who wins next week*, we have identified *the cost*, in advance, with time to act.

Cost basis: abuse is a **weekly state**. The upper bound is a perfect oracle that A-books an account exactly for the weeks it is about to be profitably toxic — the sum of every profitable Toxic/Arbitrage week (cent- and contract-normalised USD).

Start with the whole universe, and split it honestly.



- $U = B \cup C$. Every account, every day, is one or the other.
- **C (Toxic / Arbitrage)** — accounts whose *behaviour* matches one of a configurable, extensible set of *measurable* classes.
- **B** — everyone else, correctly retained in the B-book: they are not costing us.
- $A \subset C$ — the subset that actually costs money. Defined next.

The class definitions are inputs, not hard-coded. Any operator can add, edit or re-weight them — the framework optimises around whatever is agreed.

Abuse = Toxic behaviour $\cap$ money made in the next n days.

Being in C is a behaviour. It only becomes a **cost** when it makes money at our expense. So we intersect C with future profitability:

$$A = C \cap \{ \text{account gains equity over the next } n = 5 \text{ days} \} \quad (n \text{ configurable \& optimised})$$

Why this is the right definition

- **Abuse is a weekly state** — an account abusive this week can be harmless the next. So we evaluate it every window, not once for life.
- We act only on **A**, never all of C — harmless "toxic-looking" flow stays in the B-book earning revenue. False positives are expensive too.
- The benchmark is a **perfect oracle** that A-books an account exactly for its profitable toxic weeks — the natural 100% upper bound we measure against.

Why balance & deposits don't enter

A client could withdraw the gain at the end of day 5. Only **money made after the qualifying behaviour** is the adverse cost — so the measure is a clean equity delta, consistent across cent accounts and contract specs.

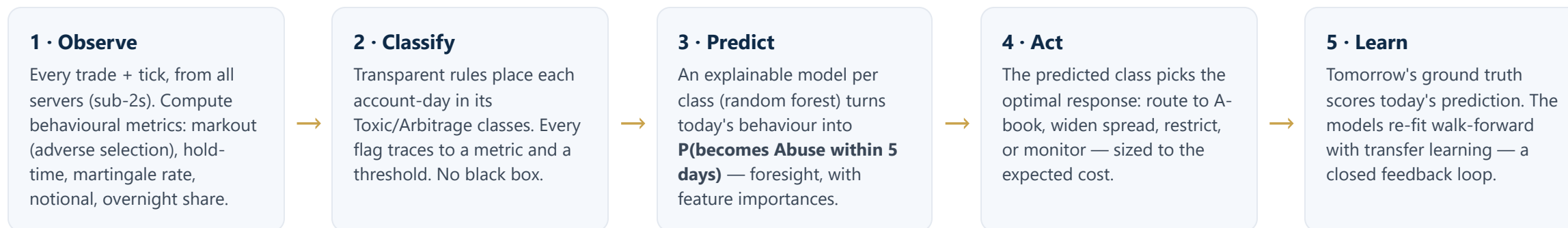
Six classes today, ranked by the money at stake.

TOXIC / ARBITRAGE CLASS	ABUSE ACCOUNTS (A)	ABUSE-WEEKS	COST / DAY (USD)	MODEL AUC	RELATIVE IMPACT
Persistent edge	12,853	46,383	\$384,657	0.891	
High magnitude	3,996	8,901	\$374,660	0.865	
Scalper	10,517	32,111	\$292,688	0.751	
High-exposure / recovery	781	1,511	\$179,337	0.952	
News / volatility	4,829	13,317	\$89,210	0.843	
Martingale	5,125	10,970	\$82,230	0.796	

CONFIGURABLE

Each class is a measurable rule the desk can edit; new classes are added the same way. We attack the highest-impact classes first — every class carries a different % of members that turn into Abuse.

Five steps, every one explainable to a first-year analyst.



The intelligence is powerful **because** it is simple: a chain of measurable definitions, a transparent classifier, and a model whose every driver can be named. If it flags an account, we can always say **exactly why**.

Foresight, measured out-of-sample.

CLASS MODEL	HOLDOUT AUC	SKILL
High-exposure / recovery	0.952	
Persistent edge	0.891	
High magnitude	0.865	
News / volatility	0.843	
Martingale	0.796	
Scalper	0.751	

0.85

Mean out-of-sample AUC across classes

AUC is the probability the model ranks a true future-Abuser above a non-Abuser. **0.85 means it is right ~85% of the time** at telling tomorrow's cost from tomorrow's noise — today.

Every model is a random forest: inspectable, reproducible, and it names its own top drivers. No opaque deep-net.

One objective, made SMART.

Within two weeks, detect $\geq 75\%$ of next-week Abuse-USD in advance (out-of-sample) across the top-six classes, recovering $\geq \$341\text{k/day}$ ($\sim \$86\text{M/yr}$) of B-book P&L — measured daily against realised ground truth.

SMART	HOW THIS OBJECTIVE SATISFIES IT
Specific	Detect the accounts that will become Abuse (A) within 5 days, per class, and act.
Measurable	Out-of-sample detection of Abuse-USD (recall) + precision; USD recovered/day vs the walk-forward baseline.
Achievable	Models already average 0.85 AUC — comfortably above the 0.75 bar — on Zeal's own history.
Relevant	Directly protects the B-book's core P&L: \$454k/day is at stake today.
Time-bound	A 2-week live test with a clear go / no-go, reported daily.

Predicted vs actual, every day — a closed loop.

Predict (day D)

Score every non-member: P(becomes Abuse by D+5). Publish the watchlist & recommended actions.



Observe (day D+5)

New ground truth arrives: who actually became Abuse, and the USD they took.



Score

Precision, recall and USD-captured land on a live dashboard — the target is audited continuously.



Re-fit

Walk-forward transfer learning folds the new truth back in. The system compounds accuracy over time.

Self-correcting

Definitions stay fixed; the models adapt to drift automatically.

Fully auditable

Every prediction is later checked against what really happened.

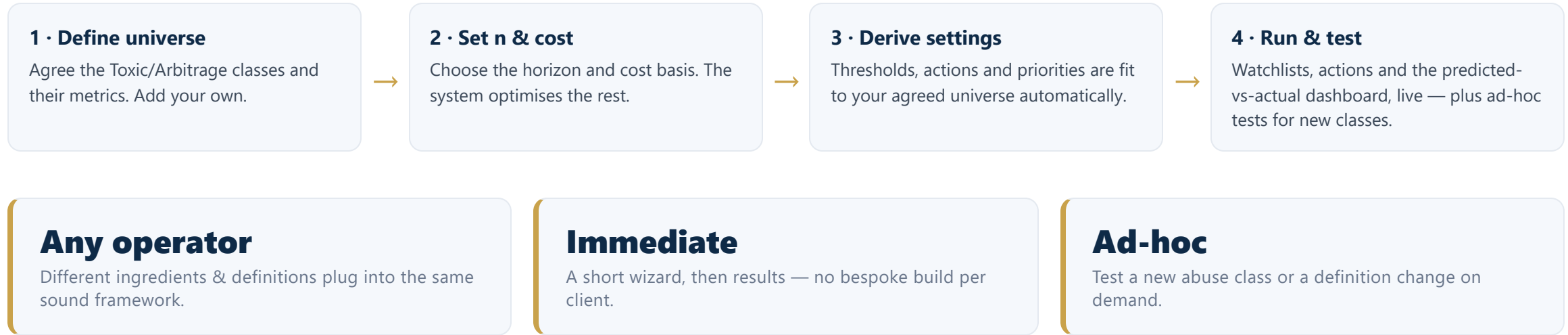
Turn the lens around: find the profit we can't yet explain.

- Search for accounts that are **profitable over the 5-day window** and **predictable in advance** — yet fit *no* existing class.
- These are candidate **new abuse classes**, surfaced automatically by mining every source: trades, ticks, funding, markout, exposure.
- If the AI can explain the pattern → we **name a new class** and add it.
- If it cannot yet → we can still **act on the evidence** (predictable adverse profitability) before the explanation matures, when the logic and the money justify it.

The framework doesn't just police the abuse we know. It is a **discovery engine** for the abuse we don't — the difference between reacting to last quarter's arbitrage and getting ahead of next quarter's.

BUILT AS A FRAMEWORK, NOT A FIXED MODEL

Vendor-agnostic by design — configure, then run.



THE ASK

A two-week live pilot on the top-six classes.

TARGET

$\geq 75\%$ detection

of next-5-day Abuse-USD, out-of-sample

PRIZE

~\$86M / year

B-book P&L recovered at target

PROOF

Daily, audited

predicted vs actual, clear go / no-go

The cost is measured. The models are built and out-of-sample. The logic is explainable end-to-end. **The only decision left is to switch it on.**